

KOGALYM, Russian Federation

WMO: 237480

Lat: 62.1975N

Lon: 74.5388E

Elev: 221

StdP: 14.58

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.8	-38.4	-47.5	0.3	-42.9	-42.5	0.5	-37.8	20.9	17.8	19.4	15.0	4.1	20	0.743

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	84.1	67.6	80.4	66.3	76.6	64.3	69.9	80.2	67.9	77.1	65.8	73.7	8.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	98.0	73.8	64.5	91.6	72.3	62.5	85.4	71.4	34.0	80.0	32.4	77.4	30.8	73.1	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.6	16.6	DB	-47.9	86.9	7.8	3.3	-53.6	89.3	-58.1	91.2	-62.5	93.1	-68.2	95.5	(4)
(5)			WB	-47.2	72.4	6.8	2.4	-52.2	74.2	-56.1	75.6	-60.0	77.0	-64.9	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	-7.6	-4.8	14.4	24.8	39.4	58.7	65.0	56.5	45.6	29.5	9.6	-0.1	(6)
(7)		DBStd	27.57	18.06	16.63	13.03	12.09	10.62	9.74	7.43	6.36	7.70	9.79	15.77	17.19	(7)
(8)		HDD50	9172	1785	1534	1104	757	367	34	1	13	176	636	1212	1553	(8)
(9)		HDD65	13739	2250	1954	1569	1207	798	232	96	272	581	1100	1662	2018	(9)
(10)		CDD50	1060	0	0	0	0	37	296	465	215	46	1	0	0	(10)
(11)		CDD65	151	0	0	0	0	3	43	95	9	1	0	0	0	(11)
(12)		CDH74	1321	0	0	0	0	37	381	837	63	3	0	0	0	(12)
(13)		CDH80	334	0	0	0	0	7	88	232	7	0	0	0	0	(13)
(14)	Wind	WSAvg	8.5	7.4	7.5	9.2	9.4	9.6	8.7	7.9	8.2	8.6	9.1	8.1	7.8	(14)
(15)	Precipitation	PrecAvg	32.4	2.1	1.7	2.3	2.1	2.5	3.1	2.8	3.5	2.9	3.4	3.3	2.6	(15)
(16)		PrecMax	40.6	4.0	3.3	5.9	3.3	5.9	5.4	6.7	5.9	5.7	5.9	5.4	5.3	(16)
(17)		PrecMin	25.7	0.6	0.4	0.1	0.1	1.3	1.4	1.2	1.3	1.1	1.5	1.2	1.1	(17)
(18)		PrecStd	3.8	0.9	0.7	1.1	0.8	1.1	0.9	1.1	1.1	1.1	1.1	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.6	34.1	40.7	52.1	78.7	87.1	89.5	79.8	71.2	55.2	38.0	31.9	(19)
MCWB			31.4	32.2	34.8	44.9	62.7	65.4	69.7	68.1	59.9	48.6	37.4	31.3	(20)	
2%		DB	26.3	28.0	36.3	45.3	70.0	82.4	85.8	75.2	64.8	48.5	34.8	29.8	(21)	
		MCWB	25.5	26.8	32.6	40.0	57.4	66.3	68.0	64.1	57.2	45.2	34.1	29.3	(22)	
5%		DB	21.6	22.2	33.4	42.5	62.9	78.6	82.4	71.7	60.7	45.4	33.1	26.2	(23)	
		MCWB	21.0	20.8	30.8	38.3	53.3	64.6	67.9	62.2	54.8	43.3	32.6	25.6	(24)	
10%		DB	17.3	18.3	31.2	39.6	55.8	74.7	78.8	68.2	57.2	41.8	30.5	22.3	(25)	
		MCWB	16.7	17.2	29.1	36.2	48.9	62.9	65.9	60.0	53.1	40.5	29.9	21.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	32.3	35.4	45.9	63.7	72.0	72.5	70.4	60.4	49.2	37.5	31.0	(27)
MCD B			30.7	34.3	39.5	51.1	76.5	80.1	83.1	77.7	66.4	53.7	37.9	31.3	(28)	
2%		WB	25.4	26.7	32.9	41.1	58.4	68.6	70.4	66.3	57.9	45.8	34.1	29.0	(29)	
		MCD B	26.0	28.0	35.4	44.1	70.0	77.2	81.3	72.9	63.5	47.8	34.3	29.4	(30)	
5%		WB	21.0	21.0	31.3	38.5	54.2	66.3	69.0	63.6	55.8	43.4	32.3	25.5	(31)	
		MCD B	21.4	22.1	33.4	41.5	61.7	73.7	79.4	69.2	60.0	45.1	32.8	26.1	(32)	
10%		WB	16.5	17.3	29.1	36.5	49.4	64.4	67.3	61.4	53.4	40.3	30.0	21.6	(33)	
		MCD B	17.1	18.3	31.0	39.5	56.0	72.0	76.7	66.7	56.9	41.6	30.4	22.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.2	16.2	16.9	16.3	15.1	16.1	16.3	14.2	11.5	9.0	12.2	13.4	(35)
MCDBR			14.5	13.6	14.5	16.2	24.1	21.3	21.9	19.9	16.5	10.5	9.9	11.1	(36)	
MCWBR			14.4	13.2	12.4	12.2	14.2	9.8	9.2	10.6	10.0	7.8	9.8	11.0	(37)	
5% WB		MCDBR	14.9	13.8	15.7	14.3	22.8	18.4	19.0	17.5	15.9	9.9	9.8	11.4	(38)	
		MCWBR	14.9	13.4	14.2	11.2	14.1	10.3	9.0	10.6	10.3	7.6	9.8	11.3	(39)	
		(40)	Clear-Sky Solar Irradiance	taub	0.246	0.291	0.344	0.365	0.378	0.394	0.422	0.409	0.354	0.320	0.249	0.233
taud	1.939	2.016		1.966	2.053	2.296	2.365	2.274	2.279	2.400	2.240	1.990	1.892	(41)		
Ebn at Noon	123	200		231	256	266	263	250	242	237	193	120	66	(42)		
Edh at Noon	12	24		37	42	36	35	37	34	25	20	11	7	(43)		
(44)	All-Sky Solar Radiation	RadAvg	98	370	889	1473	1669	1699	1610	1114	677	275	92	45	(44)	
(45)		RadStd	7	30	50	137	117	140	174	130	79	39	23	3	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-662	N/A	N/A

Nomenclature: See separate page